Manav Jain

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Education

Bachelor of Technology in Information Technology 2024 – 2028
SVKM's Dwarkadas J. Sanghvi College of Engineering, Mumbai – CGPA: 8.5/10

Skills

Languages: Python, Java, SQL, C++, C

Machine Learning / Deep Learning: PyTorch, TensorFlow/Keras, Scikit-learn, HuggingFace, Computer Vision, NLP, Transformers

LLM / Generative AI: RAG, Fine-tuning, LoRA/QLoRA, LLM Evaluation, Agentic AI & Multi-Agent Systems, Model Context Protocol, Multimodal Models, Guardrails & Safety

Frameworks & Libraries: LangChain/LangGraph/LangFuse/LangSmith, FastAPI, Streamlit, Pydantic, SHAP/Grad-CAM

Vector Databases & Data: Pinecone, ChromaDB, FAISS, PostgreSQL, Pandas, NumPy

Deployment: Docker

Cloud Platforms: AWS (SageMaker, S3, Bedrock), GCP (Vertex AI), Azure

Developer Tools & Practices: Git/GitHub, Testing (pytest), Observability

Experience

Student Researcher – Skin Disease Prediction (Innovative Product Design) Ongoing
DJ Sanghvi College of Engineering, Mumbai

- Trained an ensemble of Swin-T, ConvNeXt, and EfficientNet-B1 on 14,500 dermatological images of Indian skin tones, achieving 86% accuracy and 0.978 AUC-ROC, surpassing standard medical benchmarks.
- Implemented and tested a federated learning setup with differential privacy to enable multi-institution model training without centralizing patient data, addressing data-sharing constraints under medical privacy standards.

Research Member – Adversarial Robustness Under Compression for Edge ViTs GitHub Ongoing
Google Developer Group (GDG), DJ Sanghvi College of Engineering, Mumbai

- Designed a clean/adversarial batch-mixing adversarial training scheme that cut attack success rate by over 50% relative (FGSM ASR 38.6%→16.4%, Patch ASR 70%→59.6% at INT8), eliminating the overfitting seen in single-step AT.
- Knowledge distillation beat adversarial training on quantized models (Patch ASR 67.6%→59.6% at INT8); quantization alone nearly doubled attack success despite under 5% accuracy loss.

Projects

SkinSure: Skin Disease Classification for Fitzpatrick IV–VI GitHub Feb 25

- Built an end-to-end preprocessing pipeline for dermatological images (Fitzpatrick IV–VI) with automated hair removal, artifact elimination, and color normalization to standardize inputs for underserved darker skin tones, then integrated Grad-CAM and SHAP explainability into inference for clinician-readable justifications.
- Designed an agentic clinical workflow layer that orchestrates model output into severity triage, clinical summarization, and referral/pharmacy routing, automating the path from raw prediction to actionable next step without manual clinician handoff.

CHAKRA – Critical Infrastructure Cyber-Risk Platform GitHub Oct 25

- Achieved a 27.3× and 130× lift over baseline (PR-AUC 0.217, 0.348) with 3 ML models (IsolationForest, RandomForest, XGBoost) for ICS/SCADA anomaly detection, gated by 12/12 automated checks.
- Converted live model scores into risk multipliers in 1.2–1.4ms/event for a real-time Monte Carlo engine, paired with a guardrailed LLM explainer that strips any ungrounded number.
- Containerized the dashboard, API, and precomputed cache into a single Docker image, enabling one-command deployment via Docker Compose.

Vidya – GenAI EdTech Tutoring Assistant GitHub Mar 26

- Designed an adaptive Persona Engine with an SQLite-based memory system that tracks student confusion points across sessions to switch between Tutor, Buddy, and Mentor roles.
- Engineered a multimodal Retrieval-Augmented Generation pipeline using LangChain and FAISS to ground responses in academic textbooks, reducing hallucinations and providing 100% verifiable study references.

Manuscripts

- M. Jain*, D. Bhagchandani, M. A. Fatteh, R. Dhand, N. Katre. "Ensemble Learning for Early Detection of Skin Disease in Indian Skin Tones." Manuscript, 2026.
- M. Jain*, S. Shah, P. Khandagle, N. Katre, S. Machado. "Adversarial Robustness of Compressed Vision Transformers for Edge Deployment." Manuscript, 2026.

*First author.